Porovnávanie prístupov plánovania pomocou symetrickej diferencie

Pre štyri úlohy vygenerujem pomocou každého prístupu 3 riešenia a pomocou *symetric_difference.py* vyrátam normalizovanú symetrickú diferenciu medzi akciami v jednotlivých riešeniach pre každý prístup. Normalizovaná symetrická diferencia je vyjadrená v percentách – 0% = riešenia sú úplne rovnaké, 100% = riešenia neobsahujú ani jednu rovnakú akciu.

Jednotlivé prístupy sú:

1. Hopper (doména HopperDomain.pddl)
2. HopperQuest (doména HopperQuestDomain.pddl)
3. HopperCost (doména HopperCostDomain.pddl)

1. Úloha – Prines starú knihu

Hopper:

Pre plány z tejto domény potrebujem odstrániť čísla na konci akcií, aby nespôsobili, že rovnaké akcie na inom poradí sa budú pokladať za rôzne akcie.

1. riešenie
(go hopper mill village o_go n0 n1)
(go hopper village shop o_go n1 n2)
(steal hopper shopkeeper old_book shop o_steal n2 n3)
(go hopper shop village o_go n3 n4)
(go hopper village mill o_go n4 n5)
(buy mage hopper gold_coin old_book mill o_buy n5 n6)

2. riešenie
(go hopper mill village o_go n0 n1)
(go hopper village shop o_go n1 n2)
(steal hopper shopkeeper old_book shop o_steal n2 n3)
(go hopper shop village o_go n3 n4)
(go hopper village mill o_go n4 n5)
(attack_character hopper mage mill o_attack_ch n5 n6)
(buy mage hopper gold_coin old_book mill o_buy n6 n7)

3. riešenie
(go hopper mill village o_go n0 n1)
(go hopper village shop o_go n1 n2)
(steal hopper shopkeeper old_book shop o_steal n2 n3)
(go hopper shop village o_go n3 n4)
(go hopper village mill o_go n4 n5)
(attack_building hopper mill o_attack_b n5 n6)
(buy mage hopper gold_coin old_book mill o_buy n6 n7)

Symetrická diferencia:

- 1., 2.:
    - (attack_character hopper mage mill o_attack_ch)

- 1., 3. :
    - (attack_building hopper mill o_attack_b)
- 2., 3.:
    - (attack_character hopper mage mill o_attack_ch),
    - (attack_building hopper mill o_attack_b)

HopperQuest:

1. riešenie
(go hopper mill village)
(go hopper village shop)
(steal hopper shopkeeper old_book shop)
(go hopper shop village)
(go hopper village mill)
(buy mage hopper gold_coin old_book mill)

2. riešenie
(go hopper mill village)
(go hopper village meadow)
(collect hopper black_lily meadow)
(go hopper meadow village)
(go hopper village shop)
(buy shopkeeper hopper old_book black_lily shop)
(go hopper shop village)
(go hopper village mill)
(buy mage hopper gold_coin old_book mill)

3. riešenie
(go hopper mill village)
(go hopper village shop)
(attack_character hopper shopkeeper shop)
(kill hopper shopkeeper shop)
(take hopper shopkeeper old_book shop)
(go hopper shop village)
(go hopper village mill)
(buy mage hopper gold_coin old_book mill)

Symetrická diferencia:

- 1., 2.
    - (collect hopper black_lily meadow)
    - (steal hopper shopkeeper old_book shop)
    - (go hopper meadow village)
    - (go hopper village meadow)
    - (buy shopkeeper hopper old_book black_lily shop)
- 1., 3.
    - (kill hopper shopkeeper shop)
    - (take hopper shopkeeper old_book shop)
    - (steal hopper shopkeeper old_book shop)

    - (attack_character hopper shopkeeper shop)
- 2., 3.
    - (attack_character hopper shopkeeper shop)
    - (collect hopper black_lily meadow)
    - (go hopper meadow village)
    - (go hopper village meadow)
    - (kill hopper shopkeeper shop)
    - (take hopper shopkeeper old_book shop)
    - (buy shopkeeper hopper old_book black_lily shop)

HopperCost:

1. riešenie
(go hopper mill village)
(go hopper village shop)
(steal hopper shopkeeper old_book shop)
(go hopper shop village)
(go hopper village mill)
(buy mage hopper gold_coin old_book mill)

2. riešenie
(go hopper mill village)
(go hopper village shop)
(attack_character hopper shopkeeper shop)
(kill hopper shopkeeper shop)
(take hopper shopkeeper old_book shop)
(go hopper shop village)
(go hopper village mill)
(attack_character hopper mage mill)
(player_give hopper mage old_book mill)
(kill hopper mage mill)
(take hopper mage gold_coin mill)

3. riešenie
(steal hopper mage gold_coin mill)
(go hopper mill village)
(go hopper village shop)
(steal hopper shopkeeper old_book shop)
(go hopper shop village)
(go hopper village mill)
(player_give hopper mage old_book mill)

Symetrická diferencia:

- 1., 2.
    - (take hopper mage gold_coin mill)
    - (steal hopper shopkeeper old_book shop)
    - (kill hopper mage mill)
    - (kill hopper shopkeeper shop)

    - (take hopper shopkeeper old_book shop)
    - (attack_character hopper mage mill)
    - (attack_character hopper shopkeeper shop)
    - (buy mage hopper gold_coin old_book mill)
    - (player_give hopper mage old_book mill)
- 1., 3.
    - (steal hopper mage gold_coin mill)
    - (player_give hopper mage old_book mill)
    - (buy mage hopper gold_coin old_book mill)
- 2., 3.
    - (attack_character hopper mage mill)
    - (take hopper mage gold_coin mill)
    - (steal hopper shopkeeper old_book shop)
    - (kill hopper mage mill)
    - (steal hopper mage gold_coin mill)
    - (kill hopper shopkeeper shop)
    - (take hopper shopkeeper old_book shop)
    - (attack_character hopper shopkeeper shop)

2. Úloha – Získaj informáciu

Hopper:

1. riešenie
(go hopper mill village o_go n0 n1)
(spy hopper hunter note village o_spy n1 n2)

2. riešenie
(go hopper mill village o_go n0 n1)
(talk hunter hopper note village o_talk n1 n2)

3. riešenie
(go hopper mill village o_go n0 n1)
(attack_character hopper hunter village o_attack_ch n1 n2)
(spy hopper hunter note village o_spy n2 n3)

Symetrická diferencia:

- 1., 2.
    - (talk hunter hopper note village o_talk)
    - (spy hopper hunter note village o_spy)
- 1., 3.
    - (attack_character hopper hunter village o_attack_ch)
- 2., 3.
    - (talk hunter hopper note village o_talk)
    - (spy hopper hunter note village o_spy)
    - (attack_character hopper hunter village o_attack_ch)

HopperQuest:

1. riešenie
(go hopper mill village)
(spy hopper hunter note village)

2. riešenie
(go hopper mill village)
(talk hunter hopper note village)

3. riešenie
(go hopper mill village)
(go hopper village shop)
(collect hopper list shop)
(read hopper list note)

Symetrická diferencia:

- 1., 2.
    - (talk hunter hopper note village)
    - (spy hopper hunter note village)
- 1., 3.
    - (collect hopper list shop)
    - (spy hopper hunter note village)
    - (go hopper village shop)
    - (read hopper list note)
- 2., 3.
    - (talk hunter hopper note village)
    - (collect hopper list shop)
    - (go hopper village shop)
    - (read hopper list note)

HopperCost:

1. riešenie
(go hopper mill village)
(spy hopper hunter note village)

2. riešenie
(go hopper mill village)
(talk hunter hopper note village)

3. riešenie
(go hopper mill village)
(go hopper village shop)
(collect hopper list shop)
(read hopper list note)

Symetrická diferencia:

- 1., 2.
  - (talk hunter hopper note village)
  - (spy hopper hunter note village)
- 1., 3.
  - (collect hopper list shop)
  - (spy hopper hunter note village)
  - (go hopper village shop)
  - (read hopper list note)
- 2., 3.
  - (talk hunter hopper note village)
  - (collect hopper list shop)
  - (go hopper village shop)
  - (read hopper list note)

3. Úloha – Problém s vlkolakom

Hopper:

1. riešenie
(go hopper shop village o_go n0 n1)
(go hopper village meadow o_go n1 n2)
(attack_character hopper werewolf meadow o_attack_ch n2 n3)
(kill hopper werewolf meadow o_kill n3 n4)

2. riešenie
(go hopper shop village o_go n0 n1)
(go hopper village meadow o_go n1 n2)
(attack_character hopper werewolf meadow o_attack_ch n2 n3)
(attack_character hopper werewolf meadow o_attack_ch n3 n4)
(kill hopper werewolf meadow o_kill n4 n5)

3. riešenie
(go hopper shop village o_go n0 n1)
(go hopper village meadow o_go n1 n2)
(attack_character hopper werewolf meadow o_attack_ch n2 n3)
(scare_of hopper werewolf meadow o_scare n3 n4)
(kill hopper werewolf meadow o_kill n4 n5)

Symetrická diferencia:

- 1., 2.
  - -
- 1., 3.
  - (scare_of hopper werewolf meadow o_scare)
- 2., 3.
  - (scare_of hopper werewolf meadow o_scare)

HopperQuest:

1. riešenie
(go hopper shop village)
(go hopper village meadow)
(attack_character hopper werewolf meadow)
(kill hopper werewolf meadow)

2. riešenie
(go hopper shop village)
(go hopper village meadow)
(attack_character hopper werewolf meadow)
(scare_of hopper werewolf meadow)
(run_away werewolf meadow forest)

3. riešenie
-
Symetrická differencia:

- 1., 2.
    - (run_away werewolf meadow forest)
    - (kill hopper werewolf meadow)
    - (scare_of hopper werewolf meadow)

HopperCost:

1. riešenie
(go hopper shop village)
(go hopper village meadow)
(attack_character hopper werewolf meadow)
(kill hopper werewolf meadow)

2. riešenie
(go hopper shop village)
(go hopper village meadow)
(attack_character hopper werewolf meadow)
(scare_of hopper werewolf meadow)
(run_away werewolf meadow forest)

3. riešenie
-

Symetrická diferencia:

- 1., 2.
    - (run_away werewolf meadow forest)
    - (kill hopper werewolf meadow)
    - (scare_of hopper werewolf meadow)

4. Úloha – Vylieč lovca

Hopper:

1. riešenie
(go hopper village mill o_go n0 n1)
(steal hopper mage med2 mill o_steal n1 n2)
(go hopper mill village o_go n2 n3)
(cure hopper hunter med2 village o_cure n3 n4)

2. riešenie
(go hopper village mill o_go n0 n1)
(steal hopper mage med2 mill o_steal n1 n2)
(go hopper mill village o_go n2 n3)
(attack_character hopper hunter village o_attack_ch n3 n4)
(cure hopper hunter med2 village o_cure n4 n5)

3. riešenie
(go hopper village mill o_go n0 n1)
(steal hopper mage med2 mill o_steal n1 n2)
(go hopper mill village o_go n2 n3)
(player_give hopper hunter med2 village o_player_give n3 n4)
(cure hunter hunter med2 village o_cure n4 n5)

Symetrická diferencia:

- 1., 2.
  - (attack_character hopper hunter village o_attack_ch)
- 1., 3.
  - (cure hopper hunter med2 village o_cure)
  - (player_give hopper hunter med2 village o_player_give)
  - (cure hunter hunter med2 village o_cure)
- 2., 3.
  - (cure hopper hunter med2 village o_cure)
  - (player_give hopper hunter med2 village o_player_give)
  - (attack_character hopper hunter village o_attack_ch)
  - (cure hunter hunter med2 village o_cure)

HopperQuest:

1. riešenie
(go hopper village mill)
(steal hopper mage med2 mill)
(go hopper mill village)
(cure hopper hunter med2 village)

2. riešenie
(go hopper village meadow)
(go hopper meadow forest)
(collect hopper med1 forest)

(go hopper forest meadow)
(go hopper meadow village)
(cure hopper hunter med1 village)

3. riešenie
(give hunter hopper gold village)
(go hopper village mill)
(buy mage hopper med2 gold mill)
(go hopper mill village)
(cure hopper hunter med2 village)

Symetrická diferencia:

- 1., 2.
    - (go hopper meadow village)
    - (steal hopper mage med2 mill)
    - (collect hopper med1 forest)
    - (go hopper village meadow)
    - (go hopper village mill)
    - (go hopper forest meadow)
    - (go hopper mill village)
    - (cure hopper hunter med1 village)
    - (cure hopper hunter med2 village)
    - (go hopper meadow forest)
- 1., 3.
    - (buy mage hopper med2 gold mill)
    - (give hunter hopper gold village)
    - (steal hopper mage med2 mill)
- 2., 3.
    - (go hopper meadow village)
    - (collect hopper med1 forest)
    - (go hopper village meadow)
    - (go hopper forest meadow)
    - (go hopper village mill)
    - (buy mage hopper med2 gold mill)
    - (give hunter hopper gold village)
    - (cure hopper hunter med1 village)
    - (go hopper mill village)
    - (cure hopper hunter med2 village)
    - (go hopper meadow forest)

HopperCost:

1. riešenie
(go hopper village mill)
(steal hopper mage med2 mill)
(go hopper mill village)
(cure hopper hunter med2 village)

2. riešenie
(give hunter hopper gold village)
(go hopper village mill)
(buy mage hopper med2 gold mill)
(go hopper mill village)
(cure hopper hunter med2 village)

3. riešenie
(go hopper village meadow)
(go hopper meadow forest)
(collect hopper med1 forest)
(go hopper forest meadow)
(go hopper meadow village)
(cure hopper hunter med1 village)

Symetrická diferencia:

- 1., 2.
    - (steal hopper mage med2 mill)
    - (give hunter hopper gold village)
    - (buy mage hopper med2 gold mill)
- 1., 3.
    - (collect hopper med1 forest)
    - (go hopper meadow village)
    - (cure hopper hunter med1 village)
    - (steal hopper mage med2 mill)
    - (go hopper village mill)
    - (go hopper village meadow)
    - (go hopper meadow forest)
    - (go hopper forest meadow)
    - (go hopper mill village)
    - (cure hopper hunter med2 village)
- 2., 3.
    - (collect hopper med1 forest)
    - (buy mage hopper med2 gold mill)
    - (go hopper meadow village)
    - (cure hopper hunter med1 village)
    - (cure hopper hunter med2 village)
    - (go hopper village mill)
    - (go hopper village meadow)
    - (go hopper meadow forest)
    - (go hopper forest meadow)
    - (go hopper mill village)
    - (give hunter hopper gold village)

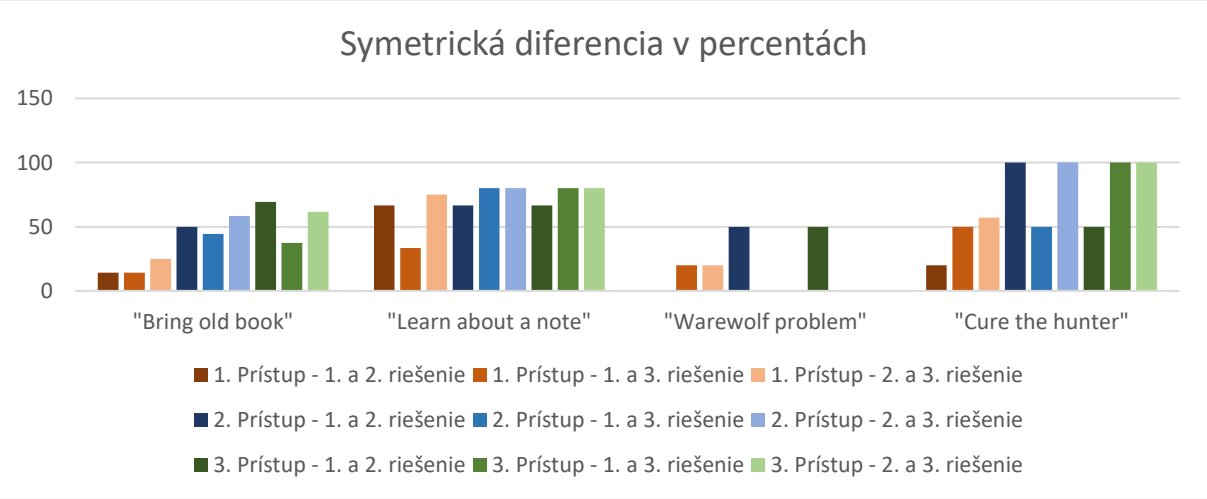
Záver

Na záver sa pozriem na normalizovanú symetrickú diferenciu medzi jednotlivými riešeniami pre jednotlivé prístupy:

Úloha	1. Prístup		
	1. Prístup - 1. a 2. riešenie	1. Prístup - 1. a 3. riešenie	1. Prístup - 2. a 3. riešenie
"Bring old book"	14,29	14,29	25
"Learn about a note"	66,67	33,33	75
"Warewolf problem"	0	20	20
"Cure the hunter"	20	50	57,14
Priemerná vzdialenosť riešení	32,977		

2. Prístup		
2. Prístup - 1. a 2. riešenie	2. Prístup - 1. a 3. riešenie	2. Prístup - 2. a 3. riešenie
50	44,44	58,33
66,67	80	80
50	-	-
100	50	100
67,944		

3. Prístup		
3. Prístup - 1. a 2. riešenie	3. Prístup - 1. a 3. riešenie	3. Prístup - 2. a 3. riešenie
69,23	37,5	61,54
66,67	80	80
50	-	-
50	100	100
69,494		



Z priemernej vzdialenosti riešení vidím, že 3. prístup vytvoril riešenia, ktoré sa od seba najviac líšia. 2. prístup je len o kúsok horší, ale prvý prístup je podstatne horší. Raz dokonca vytvoril dve identické riešenia.